ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70): scientific notation, speed, weight and arc length, Ohm's law, component symbols, resistor colour codes, binary numbers and logic gates, and workshop safety.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The SI unit of electric current is the …

A volt. B ampere. C ohm. D watt.

1.2 Which particle carries a positive charge?

A Electron B Neutron C Photon D Proton

1.3 Which energy change takes place in a dry cell?

A Chemical energy to electrical energy
B Electrical energy to chemical energy
C Heat energy to light energy
D Mechanical energy to electrical energy

1.4 A resistor has the colour bands brown, black, red. Its value is …

A 10 Ω. B 100 Ω. C 1 kΩ. D 10 kΩ.

1.5 Which instrument measures voltage?

A Voltmeter B Ammeter C Ohmmeter D Wattmeter

1.6 In a parallel circuit, the … is the same across every branch.

A current B voltage C resistance D power

1.7 Which gate gives an output of 0 when its input is 1?

A AND B OR C XOR D NOT

1.8 The binary number 0110 is equal to the decimal number …

A 4. B 5. C 6. D 7.

1.9 Which ONE of the following is a passive component?

A Resistor B Transistor C Integrated circuit D Op-amp

1.10 ESD stands for …

A electric speed device.
B electrostatic discharge.
C electron supply diode.
D energy storage device.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 The nucleus of an atom contains protons and neutrons.

2.2 Insulators have many free electrons.

2.3 A digital signal can have any value between 0 V and 5 V.

2.4 An ammeter is connected in series with the load.

2.5 The resistance of a thermistor changes with temperature.

2.6 Mass and weight are the same thing.

2.7 An antistatic wrist strap helps to prevent electrostatic damage to components.

2.8 The tolerance band shows how accurate the value of a resistor is.

2.9 An LED gives light when it is connected in reverse bias.

2.10 Energy is measured in joules.

(10)

[10]

QUESTION 3

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–M) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		**COLUMN B**
3.1	Reverse bias	A	force of gravity acting on a mass
3.2	Capacitor	B	shows the output of a logic circuit for every combination of inputs
3.3	Thermistor	C	stores electrical charge
3.4	Relay	D	variable resistor with three terminals
3.5	Proton	E	positively charged particle in the nucleus
3.6	Insulator	F	the diode blocks current because the cathode is more positive than the anode
3.7	Truth table	G	electrically operated switch
3.8	Weight	H	resistance changes with temperature
3.9	Potentiometer	I	measures voltage, current and resistance
3.10	Multimeter	J	material that does not allow current to flow easily

		K	the number of protons in an atom
		L	changes light into electrical energy
		M	rate at which work is done

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Write 0,000 47 in scientific notation. (1)

4.2 Write $3,8 \times 10^4$ in decimal notation. (1)

4.3 A delivery robot travels 2,4 km in 20 minutes. Calculate its average speed in km/h. (5)

4.4 Calculate the weight of a robot with a mass of 15 kg. Take g as 9,8 m/s². (2)

4.5 A robot wheel has a radius of 0,5 m. Calculate the length of the arc that a point on its rim travels when the wheel turns through 60°. (5)

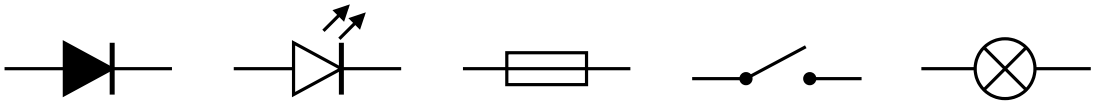
4.6 A current of 0,25 A flows through a 48 Ω resistor. Calculate the voltage across the resistor. (2)

4.7 Name FOUR forms of energy. (4)

[20]

QUESTION 5

5.1 Identify each of the following component symbols:



5.1.1 5.1.2 5.1.3 5.1.4 5.1.5

(5)

5.2 Draw the symbol of an NPN transistor and label its THREE terminals. (5)

[10]

QUESTION 6

6.1 Explain the difference between alternating current (AC) and direct current (DC). (2)

6.2 Write down the colours of the four bands of a 4,7 kΩ resistor with a tolerance of ± 5%. (4)

6.3 Give ONE use of each of the following components:

6.3.1 LDR (1)

6.3.2 Potentiometer (1)

6.3.3 Fuse (1)

6.3.4 Capacitor (1)

[10]

QUESTION 7

7.1 Convert the decimal number 13 to a binary number. Show your working. (2)

7.2 Draw the IEC symbol and the truth table of each of the following two-input gates:

7.2.1 NAND gate (5)

7.2.2 NOR gate (5)

7.3 Explain what a logic gate is. (3)

[15]

QUESTION 8

8.1 Name THREE items of personal protective equipment (PPE) used in an electronics workshop. (3)

8.2 Give THREE ways to prevent electrostatic discharge (ESD) from damaging electronic components. (3)

8.3 Give FIVE rules for the safe use of a multimeter. (5)

8.4 Differentiate between a hazard and a risk. (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER